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H- and D-atom formation from the pyrolysis of $C_6H_5CH_2Br$ and $C_6H_5CD_2Br$: Implications for high-temperature benzyl decomposition ☆

R. Sivaramakrishnan, M.-C. Su [1], J.V. Michael *

Chemical Sciences & Engineering Division, Argonne National Laboratory, Argonne, IL 60439, USA

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Abstract

The thermal decompositions of benzyl ($C_6H_5CH_2$) and benzyl,α,-d_2 ($C_6H_5CD_2$) have been studied at high temperatures with the reflected shock tube technique using H- and D-atom Atomic Resonance Absorption Spectroscopy (ARAS). The experiments were performed at high-T (1437–1801 K) at nominal pressures ≈ 0.3–1.3 atm.

The present experiments utilizing the ultra-sensitive H- and D-atom ARAS technique reveal that there are three unique channels that contribute to benzyl decomposition: H-atom removal from the ring, H/D-atom removal from the side-chain, and a non-atom producing process (postulated to be cyclopentadienyl + acetylene). The rate constants for the three channels can be represented in s^{-1} by,

$$k_{\mathrm{H,ring}} = 1.14 \times 10^{12} \exp(-32180\ \mathrm{K}/T) \quad (1475\text{–}1801\ \mathrm{K})$$

$$k_{\mathrm{H/D,side\text{-}chain}} = 4.50 \times 10^{13} \exp(-40650\ \mathrm{K}/T) \quad (1487\text{–}1789\ \mathrm{K})$$

$$k_{\mathrm{non\text{-}atom}} = 1.55 \times 109 \exp(-23470\ \mathrm{K}/T) \quad (1437\text{–}1627\ K)$$

The experimental data was also used to obtain rate constants for total benzyl decomposition and an Arrhenius fit gives,

$$k_{\mathrm{total}} = 1.05 \times 10^{12} \exp(-31450\mathrm{K}/T) \quad (1437\text{–}1801\ \mathrm{K})$$

The present study gives reliable determinations of rate constants for the three channels that, when coupled with the postulated mechanistic interpretations for these processes, should be useful in future theoretical and experimental characterizations of benzyl decomposition.
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* Corresponding author. Address: Bldg. 200, Rm. D-193, Argonne National Laboratory Argonne, IL 60439, USA. Fax: +630 252 9570.
E-mail address: jmichael@anl.gov (J.V. Michael).
[1] Special Term Appointment, Argonne National Laboratory, Argonne, IL 60439, USA. Permanent address: Department of Chemistry, Sonoma State University, 1801 E. Cotati Avenue., Rohnert Park, CA 94928, USA.

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1. Introduction

Benzyl radicals form as a product of the dominant C–H and C–C bond fission channels in the high-temperature thermal decompositions of mono-substituted aromatics (toluene, ethylbenzene, propylbenzene, and butylbenzene). These molecules are primary aromatic surrogates for gasoline, diesel, and jet fuel. Because of resonance stabilization, the benzyl radical is expected to build up to significant concentrations under typical combustion conditions. Consequently an understanding of benzyl decomposition chemistry is essential for describing the intermediates that form during pyrolysis and fuel rich oxidation of aromatic fuel surrogates. Furthermore, bimolecular reactions of benzyl with dominant combustion intermediates such as C_2H_2 and C_3H_3 are thought to play an important role in ring expansion and soot precursor chemistry [1].

The thermal decomposition of benzyl has been studied with a variety of experimental techniques. Rao and Skinner [2] utilized H and D-atom ARAS to monitor benzyl decomposition using fully and partially deuterated ethylbenzenes. Their benzyl decay mechanism involved ring opening to form $C_5H_5 + C_2H_2$ and $C_4H_4 + C_3H_3$ products in accord with the scheme proposed by Smith [3]. Single pulse shock-tube studies on toluene pyrolysis [4,5] also favor an overall mechanism for benzyl decomposition that eventually forms cyclopentadienyl + C_2H_2 and $C_4H_4 + C_3H_3$. Alternatively, shock-tube UV-absorption studies by Troe and co-workers [6,7], shock-tube H-atom ARAS studies by Braun-Unkhoff et al. [8], and photo-induced decomposition studies with molecular beam mass spectrometry by Frochtenicht et al. [9] postulate a benzyl reaction scheme that forms a C_7H_6 fragment and an H-atom. Recent shock-tube UV-absorption studies by Oehlschlaeger et al. [10] also favor this pathway. Theoretical studies by Jones et al. [11], Cavallotti et al. [12], and Da Silva et al. [13] highlight the complexity of benzyl decomposition. Two recent theoretical studies [12,13] calculate vastly different rate constants for the rate determining step, benzyl forming fulvenallene and H.

Previous H(D)-atom ARAS studies [2,8] are complicated by secondary reactions of H(D) atoms with benzyl and its products. Consequently, a primary goal of the present study is to utilize the ultra-sensitive H(D)-atom atomic resonance absorption spectrometry (ARAS) technique developed in this laboratory [14,15] to determine absolute H(D) atom yields. The data will be used to obtain branching ratios and dissociation rate constants. Furthermore H and D-atom ARAS experiments on the selectively deuterated species ($C_6H_5CD_2$) will serve to support the mechanistic aspects of benzyl decomposition, i. e. the fulvenallene route postulated by Benson [16] which is the subject of two recent theoretical studies [12,13].

2. Experimental

The present experiments were performed with the reflected shock-tube technique using H(D)-atom ARAS detection. The methods and the apparatus currently being used have been previously described [14,15] and only a brief description of the experiment will be presented here.

The shock-tube was constructed entirely from a 7-m (10.2 cm o.d.) 304 stainless steel tube with the cylindrical section being separated from the He driver chamber by a 4 mil unscored 1100-H18 aluminum diaphragm. The tube was routinely pumped between experiments to less than 1.3×10^{-11} bar by an Edwards Vacuum Products Model CR100P packaged pumping system. Shock-wave velocities were measured with eight equally spaced pressure transducers (PCB Piezotronics, Inc., Model 113A21) mounted along the downstream part of the test section and recorded with a 4094C Nicolet digital oscilloscope. Temperature and density in the reflected shock-wave regime were calculated from this velocity. This procedure has been given previously, and corrections for boundary layer perturbations have been applied [17–19]. The oscilloscope was triggered by pulses derived from the last velocity gauge signal. The photometer system was radially located at 6 cm from the endplate.

For H(D)-atom detection, the lenses were crystalline MgF2, and the resonance lamp beam intensity (filtered through 6 cm of dry air (21% O_2) to isolate the Lyman-α_H or Lyman-α_D wavelengths at 121.6 nm), was measured by an EMR G14 solar blind photomultiplier tube, as described previously [20–23], and was recorded with a LeCroy model LC334A oscilloscope. In order to measure the fraction of non-Lyman-α_H present in the resonance absorption emission lamp, an H_2 discharge flow system was used to create large [H] between the lamp and shock tube lens [22], thereby removing all of the Lyman-α_H lamp emission. The H-atom experiments were then performed with

the discharge system turned off. However, the D-atom experiments required metering very small amounts of D_2 into the resonance lamp such that the lamp intensity was similar to that for H-atoms. This ensures that the D-atom lamp will then be effectively unreversed; i. e., a Gaussian distribution [20]. In this case, D-atoms in the presence of H-atoms can be directly detected by carrying out the experiment with the H_2 discharge flow system turned on (i. e., removing Lyman-α_H) during the D-atom experiment.

3. Gases

High purity He (99.995%), used as the driver gas, was from AGA Gases. Research grade Kr (99.999%), the diluent gas in reactant mixtures, was from Praxair, Inc. The ~10 ppm impurities ($N_2 < 5$ ppm, $O_2 < 2$ ppm, Ar < 1 ppm, $CO_2 <$ 0.5 ppm, $H_2 < 1$ ppm, $H_2O < 3$ ppm, Xe < 2 ppm, and THC < 0.2 ppm) are all either inert or in sufficiently low concentration so as to not perturb H-atom profiles. For H(D)-atom detection, the microwave driven resonance lamp operated at 35 watts and 1.9 Torr of ultra-high purity He (99.999%) (effective Doppler temperature, 470 K, [24]). This grade of He contains a trace of hydrogenous impurities that are sufficient to give measurable Lyman-α_H radiation. $C_6H_5CH_2Br$ and $C_6H_5CD_2Br$ (reagent grade, 98%) were obtained from Sigma Aldrich Inc. with the latter being 98 atom % D. The purity of the liquid isotopically substituted benzyl bromide ($C_6H_5CD_2Br$) was confirmed by NMR to match the purity stated by the manufacturer, thereby negating any possibility of isotope-scrambling in the liquid phase. Both compounds were further purified by bulb-to-bulb distillation, retaining only middle thirds for mixture preparation. Using benzyliodide as the benzyl source enhances wall-absorption complications since it has a low saturation vapor pressure ~ 0.22 torr at 298 K and a melting-point of 299 K [25]. This probably necessitated the I-ARAS experiments in the Braun-Unkhoff et al. [8] study. Benzylbromide on the other hand melts at 272 K and has a higher room-T (298 K) saturation vapor pressure, ~0.54 torr [25]. Gas mixtures were accurately prepared from pressure measurements staying well below the saturation vapor pressure using a Baratron capacitance manometer in an all glass high-purity vacuum line. The internal walls of the shock-tube are polished with a mirror finish and evacuated to $<10^{-11}$ bar before each experiment. Mixtures are filled and shocks fired within 15–20 s thereby ensuring that there are no wall-absorption effects. Using similar techniques the thermal decomposition of C_6H_5I, a molecule similar to C_7H_7Br with a low saturation vapor pressure ~1.0 torr at 298 K was studied in earlier work from this laboratory [26]. I-atom yields in that study [26] (unlike the Frank et al. I-ARAS study on C_6H_5I [27]) were 1.01 ± 0.1 giving us confidence in our mixture integrities and preparation techniques.

4. Results

In the present experiments, the thermal decompositions of both $C_6H_5CH_2Br$ and $C_6H_5CD_2Br$ were used to prepare benzyl-radicals. The C–Br bond strengths are 50.5 ± 2 kcal mole^{-1} [28], and thermal decomposition of the benzyl-bromides are then expected to be nearly instantaneous for the conditions of the present experiments [11]. Jones et al. [11] have also demonstrated that benzyl bromide decomposition is a simple C-Br bond fission with no competing reactions, and, therefore, with these precursors, benzyl-radicals are instantaneously formed with concentrations equal to that of the benzyl-bromides used for each experiment.

Benzyl-radical decomposition rate constants were then measured using both D- and H-atom ARAS. A typical D-atom concentration profile from the decomposition of $C_6H_5CD_2Br$ at $T = 1673$ K is shown in Fig. 1. The simulated results using the mechanism given in Table S1 (Supplementary data) are shown as the solid black line. The long dashed lines show variations in the overall decomposition rate constant, k_{total}, by ±40%, and the short dashed lines show variations of ±0.02 in the branching ratio giving D-atoms, $BR_D = k_D/k_{total}$.

In the absence of reactions that deplete [D], these results will then be a direct measure of branching ratios. At the extreme sensitivity of the present experiment, secondary reactions are negligible contributors, as easily seen from the

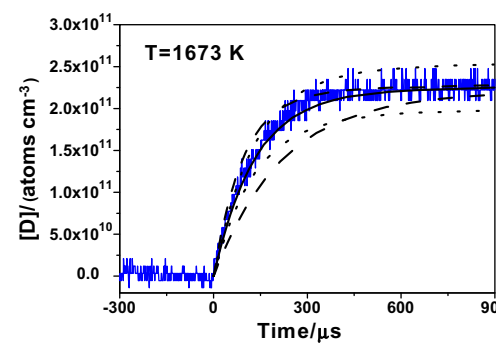

Fig. 1. [D] Profile at 1673 K. Solid line – fit using the mechanism in Table S1 with $k_{total} = 7500$ s^{-1} and $BR_D = 0.16$, dashed lines – $k_{total} \pm 40\%$ with $BR_D = 0.16$, dotted lines – $BR_D \pm 0.02$ with $k_{total} = 7500$ s^{-1}. The conditions for the experiment are $P_1 = 15.94$ Torr, $M_s = 2.603$, $\rho_5 = 3.164 \times 10^{18}$ molecule cm^{-3}, $[C_6H_5CD_2Br]_0 = 1.416 \times 10^{12}$ molecule cm^{-3}.

$[D]_\infty$ value in Fig. 1. Even with a D-atom abstraction and/or addition rate constant of 2×10^{-10} cm^3 $molecule^{-1}$ s^{-1}, the effective first-order decay constant with $[D]_\infty = 2.2 \times 10^{11}$ atoms cm^{-3} would be 44 s^{-1}, which is negligible when compared to a total formation rate of 7500 s^{-1}. This issue will be specifically addressed below where sensitivity analyses will be given. The conclusion is therefore that secondary reactions involving H or D are completely unimportant since the detectability level is so low, indicating that first-order build-up analysis holds. Hence, the present atomic formation results are a direct measure of dissociation. For D-atom experiments like Fig. 1, first-order analysis gives the simple closed form result

$$[D]_t = \{k_D[C_6H_5CD_2]_0/k_{total}\} \times \{1 - \exp(-k_{total}t)\} \quad (1)$$

where $[D]_\infty = k_D[C_6H_5CD_2]_0/k_{total}$ and $BR_D = k_D/k_{total}$. Seventeen D-atom experiments have been performed between 1517 and 1789 K at densities, ≈ 2, 3, and 6×10^{18} molecule cm^{-3}, and the conditions and results for k_{total}, k_D, and BR_D are given in Table S2 (Supplementary data). For the experiment of Fig. 1, $[C_6H_5CD_2]_0 = 1.416 \times 10^{12}$ molecule cm^{-3} giving $BR_D = 2.20 \times 10^{11}/1.416 \times 10^{12} = 0.16$. With Eq. (1) analysis, the value determined for $k_{total} = 7500$ s^{-1}, implying that $k_D = 1200$ s^{-1} as shown in Table S2.

With H-atom analysis in $C_6H_5CD_2$ decomposition experiments, the first-order formation follows

$$[H]_t = \{k_H[C_6H_5CD_2]_0/k_{total}\} \times \{1 - \exp(-k_{total}t)\} \quad (2)$$

A typical H-atom concentration profile at $T = 1594$ K is shown in Fig. 2, and the simulated $[H]_t$ from the mechanism of Table S1 is shown as the solid black line. The long dashed lines show variations in the overall decomposition rate constant, k_{total}, by ±30%, and the short dashed lines show variations of ±0.07 in the branching ratio giving H-atoms, $BR_H = k_H/k_{total}$. Fig. 3 shows a sensitivity analysis of the Fig. 2 profile indicating that only three processes contribute to H-atom sensitivity, namely

$$C_6H_5CD_2(+Kr) \rightarrow C_6H_5CD + D(+Kr), k_D \quad (3)$$
$$\rightarrow C_6H_4CD_2 + H(+Kr), k_H \quad (4)$$
$$\rightarrow \text{non-atom producing products}(+Kr), k_{NA} \quad (5)$$

This demonstration reiterates the point made above that the present detection sensitivity is so high that secondary reactions play no role in observed profiles and therefore corroborates first-order analysis. These H-atom results along with the previously described D-atom results indicate that non-atom producing products must also be formed in the decomposition. Hence, $k_{total} = k_D + k_H + k_{NA}$. Twenty H-atom experiments have been performed between 1475 and 1801 K at densities, ≈1, 2, 3, and 6×10^{18} molecule cm^{-3}, and the conditions and results for k_H and BR_H are given in Table S3. For the experiment of Fig. 2, $[C_6H_5CD_2]_0 = 1.376$ x10^{12} molecule cm^{-3} giving $BR_H = 9.57 \times 10^{11}/1.376 \times 10^{12} = 0.70$. With Eq. (2) analysis, the value determined for $k_{total} = 3200$ s^{-1}, implying that $k_H = 2240$ s^{-1} as shown in Table S3.

Lastly, rate constants for [H] formation from protonated benzyl-radicals (i.e., $C_6H_5CH_2$) have been measured between 1437 and 1627 K, and Fig. 4 shows a typical profile at 1573 K. Sensitivity analysis for the present low levels of [H] again show that first-order analysis is appropriate. The simulated $[H]_t$ from the mechanism of Table S1 is shown in Fig. 4 as the solid black line, and this is identical to that given from first-order analysis. The long dashed lines show variations in the overall decomposition rate constant, k_{total}, by ±30%, and the short dashed lines show variations of ±0.08 in the branching ratio giving H-atoms. As indicated above, three decomposition processes contribute to benzyl decay, only in this case,

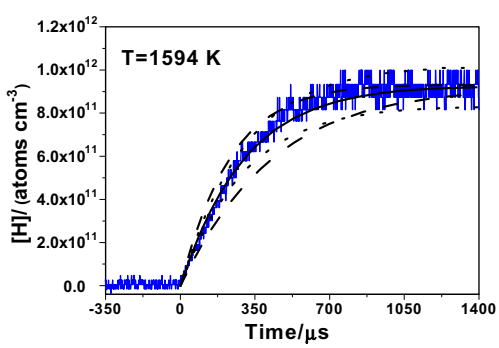

Fig. 2. [H] profile at 1594 K. Solid line – fit using the mechanism in Table S1 with $k_{total} = 3200$ s^{-1} and $BR_H = 0.70$, dashed lines – $k_{total} \pm 30\%$ with $BR_H = 0.70$, dotted lines – $BR_H \pm 0.07$ with $k_{total} = 3200$ s^{-1}. The conditions for the experiment are $P_1 = 15.86$ Torr, $M_s = 2.534$, $\rho_5 = 3.074 \times 10^{18}$ molecule cm^{-3}, $[C_6H_5CD_2Br]_0 = 1.376 \times 10^{12}$ molecule cm^{-3}.

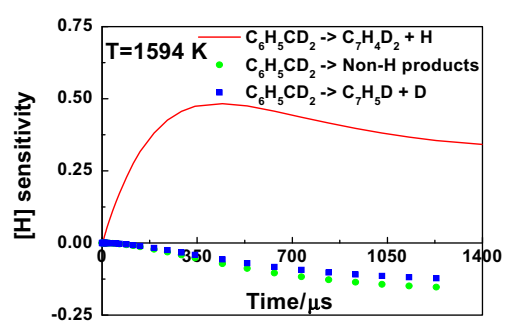

Fig. 3. H-atom sensitivity analysis for the profile in Fig. 2 using the Table S1 mechanism. The three most sensitive reactions are shown in the inset.

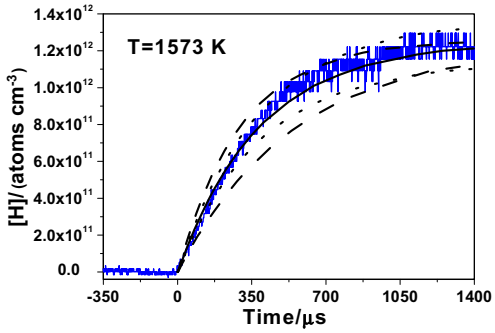

Fig. 4. [H] Profile at 1573 K. Solid line – fit using the mechanism in Table S1 with $k_{total} = 2500\ s^{-1}$ and $BR_H = 0.84$, dashed lines – $k_{total} \pm 30\%$ with $BR_H = 0.84$, dotted lines – $BR_H \pm 0.08$ with $k_{total} = 2500\ s^{-1}$. The conditions for the experiment are $P_1 = 15.93$ Torr, $M_s = 2.515$, $\rho_5 = 3.067 \times 10^{18}$ molecule cm^{-3}, $[C_7H_7Br]_0 = 1.585 \times 10^{12}$ molecule cm^{-3}.

BR_H would be derived from the sum of reactions (3) and (4) divided by k_{total}. For the experiment shown in Fig. 4 (with conditions and results for this experiment and eight other similar experiments listed in Table S4), the value for $[C_6H_5CH_2]_0 = 1.585 \times 10^{12}$ molecule cm^{-3} giving $BR_H = 1.33 \times 10^{12}/1.585 \times 10^{12} = 0.84$. This implies $BR_{NA} = 0.16$ for this experiment. Table S4 summarizes the BR_{NA} and k_{NA} derived from these experiments. Presuming that the isotope-effect for H- and D-atom dissociation from the two benzyl radicals is small since activation energies are large, the sum of the contributions from k_D and k_H from $C_6H_5CD_2$ can be identified with k_H in $C_6H_5CH_2$, and we find that the results are indeed correlated within the error of the experiments.

5. Discussion

Recent theoretical studies [12,13] (which have attempted to characterize H-atom formation from $C_6H_5CH_2$) show benzyl isomerizing to multiple intermediates with fairly low activation energies, and these isomers eventually give H and a C_7H_6 fragment (identified from theory to be fulvenallene). Since theory suggests multiple isomerization steps followed by dissociation to H, the implication, even with ultra-sensitive [H] detection, is that an unambiguous mechanistic interpretation of H formation from benzyl decomposition cannot be made. Rather the three sets of experiments described in the earlier section permit overall rate parameters to be determined for the three benzyl dissociation processes reactions (3)–(5).

Arrhenius plots of the rate constants from Tables S2–S4 are shown in Figs. 5–7, respectively. It is evident from these first order plots that there is no discernible pressure dependence for the three channels, and, in units of s^{-1}, these data can be represented as,

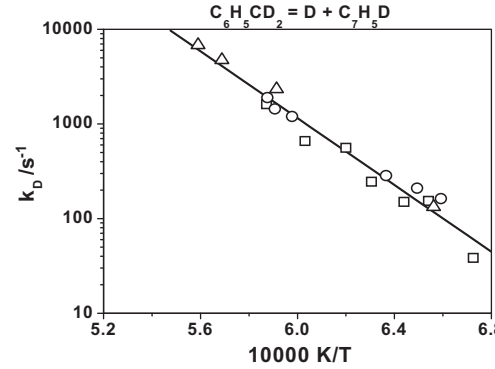

Fig. 5. Arrhenius plot for k_D. [□] – $\rho_5 \sim 2 \times 10^{18}$ molecule cm^{-3}, [○] – $\rho_5 \sim 3 \times 10^{18}$ molecule cm^{-3}, [△] – $\rho_5 \sim 6 \times 10^{18}$ molecule cm^{-3}, [▬] – Arrhenius fit, Eq. (6).

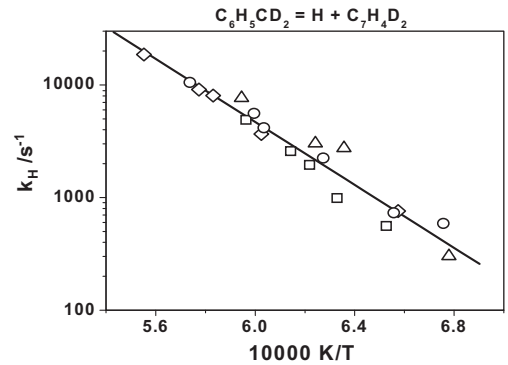

Fig. 6. Arrhenius plot for k_H. [◇] – $\rho_5 \sim 1 \times 10^{18}$ molecule cm^{-3}, [□] – $\rho_5 \sim 2 \times 10^{18}$ molecule cm^{-3}, [○] – $\rho_5 \sim 3 \times 10^{18}$ molecule cm^{-3}, [△] – $\rho_5 \sim 6 \times 10^{18}$ molecule cm^{-3}, [▬] – Arrhenius fit, Eq. (7).

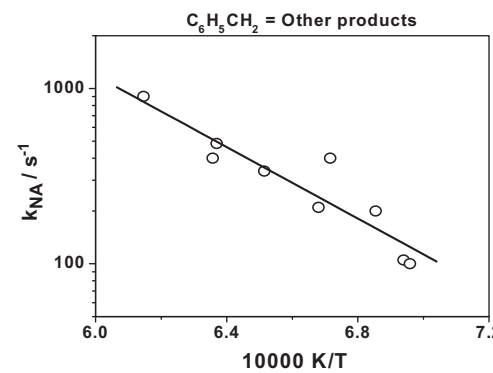

Fig. 7. Arrhenius plot for k_{NA}. [○] – $\rho_5 \sim 3 \times 10^{18}$ molecule cm^{-3}, [▬] – Arrhenius fit, Eq. (8).

$$k_D = 4.50 \times 10^{13} \exp(-40650\,K/T) \quad (1487–1789\,K) \quad (6)$$

$$k_H = 1.14 \times 10^{12} \exp(-32180\,K/T) \quad (1475–1801\,K) \quad (7)$$

$$k_{NA} = 1.55 \times 10^{9} \exp(-23470\,K/T) \quad (1437–1627\,K) \quad (8)$$

over the indicated T-ranges. The sum of k_D and k_H obtained unambiguously from the $C_6H_5CD_2$ experiments are in excellent agreement with k_H obtained from the $C_6H_5CH_2$ experiments. Figure 8 shows the H- and D-atom branching ratios as dashed and dotted lines, respectively, from the $C_6H_5CD_2$ experiments and the net H-atom branching ratio as a solid line from the $C_6H_5CH_2$ experiments. Within experimental error, the sum of the H-and D-atom branching ratios from $C_6H_5CD_2$ is in good agreement with the net H-atom branching ratios from $C_6H_5CH_2$. This is in qualitatively good agreement with the results of Rao and Skinner [2]. However, they observed a net production of 1.2 ± 0.2 H- and D-atoms from their deuterated ethylbenzene experiments in contrast to the $\approx$0.8 atoms of H + D observed in this work. The difference is no doubt due to larger initial ethylbenzene (as the benzyl source) concentrations that then lead to H and/or D-atoms produced from secondary products. The observation of D-atoms, though a small fraction $\approx$0.2, is in itself interesting in that it suggests a sequence of steps that lead to H removal from the CH_2 side-chain in benzyl in contrast to prior studies [2,12,13] that consider only ring H-atom removal.

The total rate constants $k_{total} = k_{H,ring} + k_{D/H,sidechain} + k_{NA}$ from the three data sets (Tables S2–S4) are plotted in Fig. 9. k_{total} for the three data sets, H- and D-atom data from $C_6H_5CD_2$, and the H-atom data from $C_6H_5CH_2$, are in excellent agreement. A linear least squares fit to the data gives in units of s^{-1},

$$k_{total} = 1.05 \times 10^{12} \exp(-31450\,K/T) \times (1437–1801\,K) \quad (9)$$

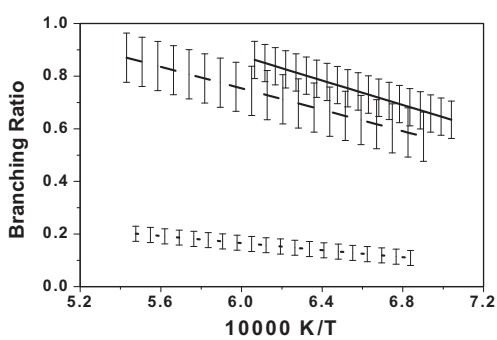

Fig. 8. Branching ratios from experiments. [▬] – fit to BR_H from C_7H_7 data, 1-standard deviation is ±0.07, [▪ ▪] – fit to BR_H from $C_6H_5CD_2$ data, 1-standard deviation is ±0.09, [▪▪▪▪] – Fit to BR_D from $C_6H_5CD_2$ data,1-standard deviation is ±0.03.

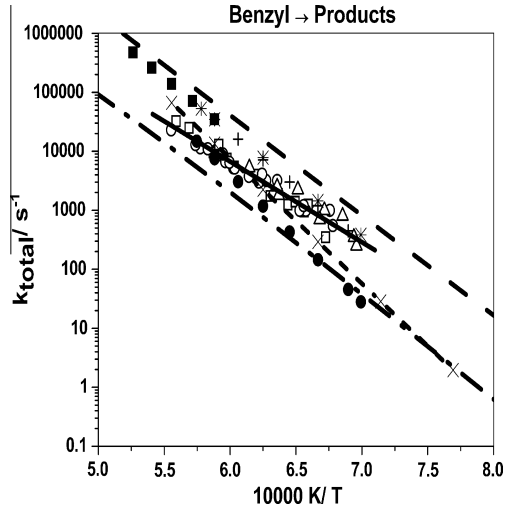

Fig. 9. Arrhenius plot for k_{total}. [□] – k_{total} from D-atom $C_6H_5CD_2$ data, [○] – k_{total} from H-atom $C_6H_5CD_2$ data, [△] – k_{total} from H-atom C_7H_7 data, [▬] – Arrhenius fit to data, Eq. (9), [- - -] – Cavallotti et al. [12], [–·–] – 1 atm, Bozzelli et al. [13], [+] – Braun-Unkhoff et al. [8], [■] – Hippler et al. [7], [*] – Oehlschlaeger et al. [10], [x- - -x] – Jones et al. [11], [●] – Rao and Skinner [2].

Equation (9) is a representation of the total rates for benzyl dissociation. Figure 9 also summarizes the available experimental and theoretical estimates for k_{total}. The present determinations are in reasonable agreement with the UV-absorption experiments of Hippler et al. [7] and Oehlschlaeger et al. [10] and the H-atom ARAS data of Braun-Unkhoff et al. [8] at $T < 1500$ K, but show increasing deviations, with the present results being a factor of 3 lower at higher-T. A likely complication with the UV-absorption experiments at higher-T could be the formation of multiple C_7H_7 isomers whose absorptions could overlap the benzyl spectra. The H-atom ARAS data [8] were performed with larger initial benzyl concentration, and, hence potential interference from bimolecular reactions cannot be ruled out. The same argument holds for the Rao and Skinner [2] H/D-atom experiments where the comparison shows that their rate constants are at least a factor of 2 lower than the present data. Interestingly the two recent theoretical efforts [12,13] predict very different rate constants over the entire T-range despite adopting the same mechanism with the rate determining step being an isomerization of benzyl to a five-membered ring intermediate. The theoretical k_∞ of Cavallotti et al. [12] is higher than the Oehlschlaeger et al. [10] measurements. On the other hand, the k_{1atm} predictions of Da Silva et al. [13] are lower that the Rao and Skinner [2] measurements. Of course, the discrepancies at high-T could also be due to pressure fall-off.

Despite the significantly different quantitative estimates for benzyl decomposition, the theoretical

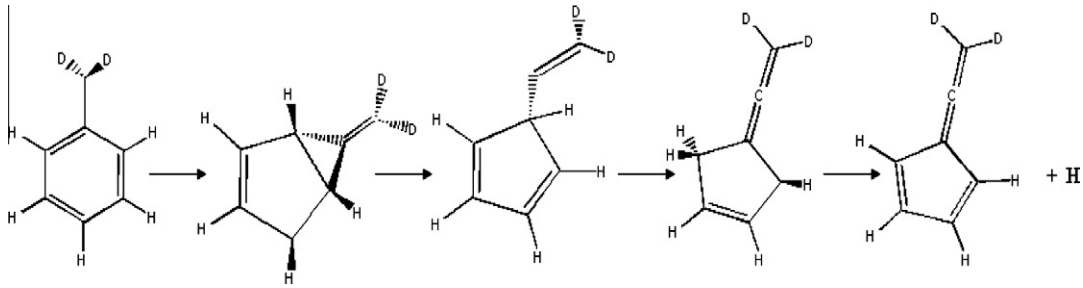

Fig. 10. Ring H-atom production scheme from benzyl decomposition.

studies [11–13] qualitatively suggest similar intermediates and transition states for the fulvenallene mechanism. We have therefore utilized the information presented in the theoretical studies [11–13] to rationalize the present data. The formation of H-atoms in the $C_6H_5CD_2$ experiments is consistent with arguments proposed [11–13,16] that suggest H-atoms come from the ring-H in benzyl. The most favored route for this H-atom production may be through a bicyclic species, 6-methylene[3.1.0] hex-3-en-2-yl (MBH), that eventually leads to H + fulvenallene, Fig. 10. The activation energy is 64 kcal/mol (Eq. (7)) for ring-H production (rate-determining step is probably isomerization of benzyl to MBH).

The formation of D-atoms from $C_6H_5CD_2$, though a minor fraction ≈0.2, suggests that there is an active route to form H atoms from the CH_2 side chain in $C_6H_5CH_2$ through a C_7H_7 isomer that is not MBH. One potential candidate could be the delocalized cycloheptatrienyl (the next most stable C_7H_7 isomer to benzyl). However the theoretical study of Cavallotti et al. [12] does not reveal a direct route leading from benzyl → cycloheptatrienyl → H + cycloheptatetraene. On the other hand, a 2,4,6-cycloheptatriene with an ortho radical centre (an intermediate in the benzyl → cycloheptatrienyl potential energy surface) can eliminate the side-chain H/D-atom, Fig. 11. This could be a candidate for the D-atoms formed from $C_6H_5CD_2$. The activation energy is 81 kcal/mol (Eq. (6)) for side-chain H production, and the rate-determining step is probably isomerization of benzyl to the bicyclic intermediate, norcaradienyl (Fig. 11).

It is clear that benzyl decomposition is a complicated issue with a variety of other C_7H_7 intermediates perhaps in equilibrium with benzyl at high-T [12]. The two schemes suggested here appear to provide plausible arguments for H and D production from $C_6H_5CD_2$. The present experiments also indicate that a minor fraction, ≈0.2, of benzyl produces non-atom products. Species analysis from toluene pyrolysis experiments [3–5] and theoretical studies [11] indicate that the most likely non-atom products are C_5H_5 (possibly cyclopentadienyl) and acetylene. These products can form from the dissociation or recombination of products/fragments of benzyl decomposition [12] or from any one of the numerous C_7H_7 intermediates. Energetic arguments favor cycloheptatrienyl to be a likely C_7H_7 intermediate that can form these products by isomerization to a bicyclic intermediate followed by rapid dissociation [29]. The activation energy of about 47 kcal/mol (Eq. (8)) and the low A-factor, suggest the rate-determining step is probably isomerization of benzyl to a bicyclic species.

6. Conclusions

The thermal decompositions of benzyl ($C_6H_5CH_2$) and benzyl,α,-d_2 ($C_6H_5CD_2$) have been studied at high temperatures behind reflected shock waves. The experiments were performed at high-T (1437–1801 K) at nominal pressures ≈ 0.3–1.3 atm. The present experiments utilizing the ultra-sensitive H- and D-atom ARAS technique

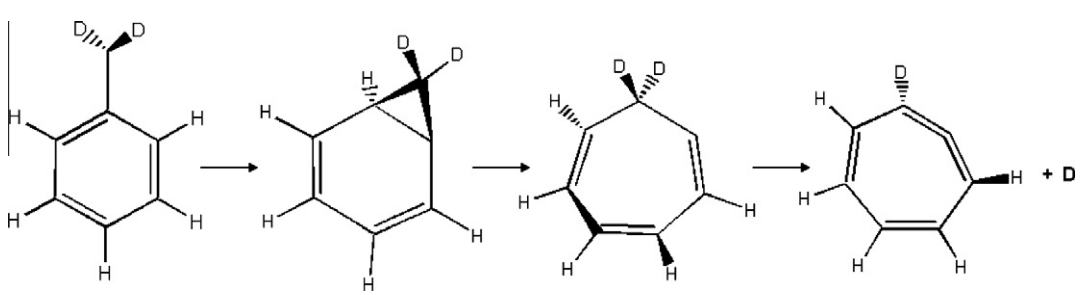

Fig. 11. Side-chain H-atom production scheme from benzyl decomposition.

reveal that there are three unique channels that contribute to benzyl decomposition, H-atom removal from the ring, H/D-atom removal from side-chain, and a non H producing process (postulated to be cyclopentadienyl + acetylene). The present study gives reliable determinations of rate constants for the three channels that, when coupled with the postulated mechanistic interpretations for these processes, should be useful in future theoretical and experimental characterizations of benzyl decomposition.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.133.

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